

Performing Reconnaissance — Learner Lab Document

Target: scanme.nmap.org

Your role: Junior Cybersecurity Analyst

Lab type: Unassisted, hands-on, evidence submission

Estimated time: 60 minutes

Companion document: *Performing Reconnaissance — Hints Companion* (file 09). If you get stuck, the companion offers hints, sample outputs, and common-mistake warnings for every task. Use of the companion is optional.

1. Lab Name

Performing Reconnaissance Against an Authorized Public Target

2. Lab Purpose

This lab introduces the first phase of any security assessment: structured information gathering. Before any vulnerability scan, penetration test, or third-party risk review can begin, an analyst must know what the target looks like from the outside — its IP, its DNS records, its registration details, its public-facing technology, and what the organization has already published about itself.

You will perform reconnaissance against a real, publicly resolvable target that has been explicitly authorized by its operator for educational use: scanme.nmap.org, a host operated by the **Nmap Project** specifically for learners to practise against. You will capture command output to files, take evidence screenshots, and produce a written reconnaissance report that explains how each finding supports vulnerability management and third-party risk review.

This is a **recon-only** lab. The Nmap Project explicitly authorizes Nmap port scanning against scanme.nmap.org for educational use; even so, this lab is scoped to passive reconnaissance only. You will not run Nmap, port scans, brute-force tools, or any active probes against the target. The reconnaissance phase is what *precedes* scanning — that is what you are practising.

3. Business Scenario

Your organization uses **Nmap** as a standard tool across its IT operations and security teams. Because Nmap is open-source third-party software, your security team performs periodic public-information reviews of the Nmap Project as part of its third-party-risk programme. The exercise has two goals: confirm that the project's public posture is consistent with what your organization expects from a security-critical dependency, and produce a baseline for future reviews.

As the junior analyst on the team, you have been assigned reconnaissance targets associated with the Nmap Project:

- scanme.nmap.org — the project's authorized testing host (used for connectivity, DNS lookups, and confirmation of the project's authorization notice).
- nmap.org — the parent registered domain (used for whois and authoritative DNS reconnaissance).

Your team lead wants you to:

1. Confirm your own analyst workstation's network identity.
2. Verify you can reach the assigned target.
3. Collect public-facing information from the Nmap Project's website.
4. Run domain registration and DNS reconnaissance against the parent domain.
5. Practice safe search-engine reconnaissance and OSINT lookup techniques using approved demo data.
6. Organize evidence into a structured submission.
7. Produce a written reconnaissance report.

You must stay strictly within the authorized scope: passive reconnaissance only. The operator's authorization to scan does **not** license you to scan in this lab — your scope is tighter than what the operator allows.

4. Skills Developed by the Lab

By completing this lab, you will practice and demonstrate:

- Capturing your own workstation's network identity using `ip`.
 - Testing reachability with `ping` and saving output to a file using shell redirection (`>` and `>>`).
 - Verifying file content with `ls` and `cat`.
 - Performing domain registration lookups using `whois`.
 - Running DNS reconnaissance in both interactive (`nslookup`) and non-interactive (`dig`) modes.
 - Reading and interpreting DNS record types: A, MX, NS, CNAME, SOA.
 - Querying authoritative name servers directly, not just the default resolver.
 - Performing structured website observation and extracting useful public information.
 - Using search-engine operators (`site:`, `filetype:`, `intitle:`, quoted phrases, the minus operator).
 - Using an approved OSINT lookup tool to fingerprint public infrastructure.
 - Recognizing OSINT exposure patterns using a safe fictional dataset.
 - Organizing reconnaissance evidence into a reviewable folder structure.
 - Writing an analyst-grade reconnaissance report that ties findings to vulnerability management and third-party risk review.
-

5. Learning Objectives

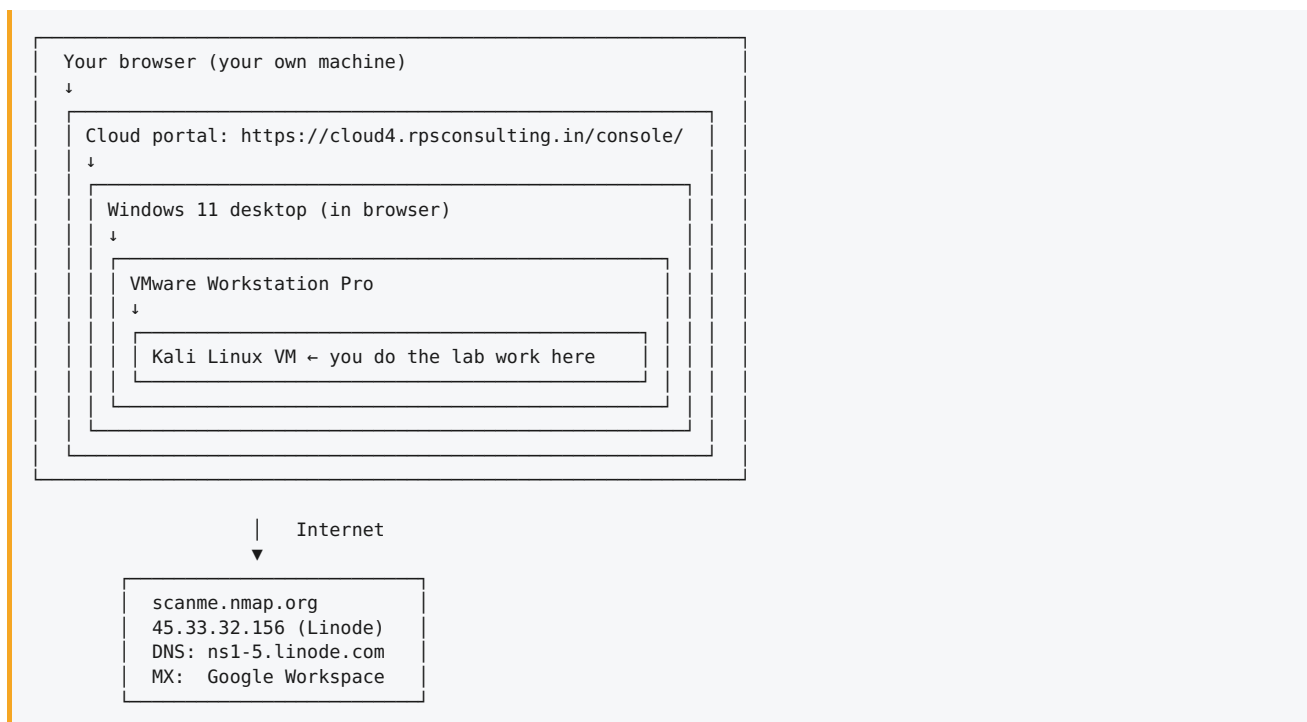
By the end of this lab, you will be able to:

1. **Identify** the workstation's network identity using the `ip` command.
 2. **Capture** connectivity data against an authorized public target and **record** it to a text file.
 3. **Query** domain registration information using `whois` and **document** the registrar, name servers, and key dates.
 4. **Analyze** DNS records using both `nslookup` (interactive) and `dig` (non-interactive), including queries against the authoritative name server.
 5. **Verify** that captured output files exist and contain the expected content using `ls` and `cat`.
 6. **Correlate** website observations, DNS findings, and OSINT lookup data into a single reconnaissance summary.
 7. **Document** findings in a structured report that explains how each finding supports vulnerability management or third-party risk review.
-

6. Lab Architecture and Requirements

6.1 Lab Topology

You work through three nested environments to reach your analyst workstation:



6.2 Target details

Item	Value
Target hostname (connectivity / DNS)	scanme.nmap.org
Project main website (Task 3 observation)	https://nmap.org
Parent domain (for whois)	nmap.org
Authoritative DNS servers	ns1.linode.com , ns2.linode.com , ns3.linode.com , ns4.linode.com , ns5.linode.com
Approved demo targets for search recon	example.com , example.org (or others on your lab portal)
Operator authorization	Published on the scanme.nmap.org welcome page

6.3 Lab requirements

On your Kali workstation:

- A terminal application (preinstalled).
- Firefox web browser (preinstalled).
- Tools: `ip` , `ping` , `ls` , `cat` , `whois` , `nslookup` , `dig` .
- A text editor (`nano` , `gedit` , or VS Code).
- A screenshot tool (`gnome-screenshot` or `Flameshot`).
- The pre-staged working folder `~/recon_lab/` containing:
 - `report_template.md` (or `.docx`) — the report you'll fill in for Task 12.
 - `demo_people_data.txt` — fictional dataset for Task 10.
 - `screenshots/` — empty folder where you'll save all evidence images.

6.4 Accessing the lab environment

You must reach your Kali workstation before doing anything else. Pass through these five steps in order:

Step 1 — Log into the cloud portal

Open a browser on your own machine and go to:

<https://cloud4.rpsconsulting.in/console/>

Sign in with the portal credentials your instructor provided.

Step 2 — Sign on to Windows 11

Once the portal loads, a Windows 11 desktop appears inside your browser.

- **Password:** `P@%w0rd`

Note the unusual password — the third and fourth characters are **two percent signs (%)**, not a typo. Type the password exactly as shown: capital P, the at-sign, two percent signs, lowercase w, the digit zero, lowercase r, lowercase d.

Step 3 — Open VMware Workstation Pro

On the Windows 11 desktop, locate **VMware Workstation Pro** (desktop shortcut or Start menu) and open it. Click through any license / evaluation dialog to reach the main window.

Step 4 — Start the Kali Linux VM

In VMware Workstation, find the **Kali Linux** entry in the VM library, then click **Power on this virtual machine**. Wait roughly one minute for the VM to boot to the login screen.

Step 5 — Log into Kali

At the Kali login screen:

- **Username:** `kali`
- **Password:** `kali`

You're ready when the Kali desktop has loaded and you can open a terminal.

6.5 Pre-flight verification

Confirm your environment is ready before starting Task 1:

- Your working folder is `~/recon_lab/`.
- The report template (`report_template.md` or `.docx`) is present.
- The fictional dataset (`demo_people_data.txt`) is present.
- The `screenshots/` folder is present and writable.
- DNS resolution and outbound connectivity to the target work.

If any of these fail, fix the issue before continuing.

7. Tasks

This is an **unassisted, hands-on lab**. Each task below tells you the **purpose**, any **constraints or scope rules**, what to **deliver**, how to **verify** you're done, what to **record**, and what **evidence** to submit. The task does not give you the commands — that's the assessment.

If you get stuck, the *Hints Companion* document offers hints, sample outputs, and common-mistake warnings for each task. Using it is optional but expected for entry-level learners.

Complete the tasks in order — each builds on the previous one.

Task 1 — Determine Your Kali System's Network Identity

Purpose: Establish your own workstation's network identity before touching anything else. Every report begins with "who am I and where am I on the network."

Output file: none.

Verification step: the screenshot clearly shows the network configuration of your `eth0` interface, including an IPv4 address.

Information to record:

- Kali IPv4 address.
- Network interface name.
- Subnet (e.g., `/24`).

Evidence to submit: `screenshots/task01_ip.png`.

Task 2 — Capture and Record Network Connectivity Data

Purpose: Verify you can reach the assigned target, and practice saving command output to a file using shell redirection.

Constraints:

- Send exactly four ICMP echo requests — not an unbounded ping.
- Save the output to a file named `target_info.txt` in `~/recon_lab/`.

Output file: `target_info.txt` (in `~/recon_lab/`).

Verification step: `target_info.txt` exists, is larger than 100 bytes, and contains ping statistics (look for "packets transmitted").

Information to record:

- Target reached (Y/N).
- Packets transmitted.
- Packets received.
- Packet loss (%).
- Average response time.
- The IP address ping resolved the hostname to.

Evidence to submit: `target_info.txt`, `screenshots/task02_ping.png`.

Task 3 — Explore the Target Website Through Firefox

Purpose: Practice structured website observation. Public-facing pages often contain more reconnaissance value than command-line tools.

Required coverage:

- Browse the Nmap Project's main website at `https://nmap.org` and observe at least four distinct pages (for example: home, about, download, documentation, reference guide).
- Visit `http://scanme.nmap.org` and read the operator's authorization notice in full.

Scope reminders: - Browse only. Do **not** download any installers or binaries. - Do **not** run Nmap or any other scanner against `scanme.nmap.org` or any Nmap Project asset, even though the operator permits Nmap scanning. This lab is scoped to the reconnaissance phase that precedes scanning.

Output file: none.

Verification step: screenshots clearly show the target URL in the address bar and the relevant page content.

Information to record:

- Project / site display name (as it appears in the page header or title).
- Page types observed on `nmap.org` (About, Download, Docs, Reference Guide, etc.).
- Visible technologies hinted at on the site (URL patterns, footer notes, page generators).
- Any visible contact information.
- A summary of the authorization notice published on `scanme.nmap.org`.
- Any other interesting public information.

Evidence to submit: `screenshots/task03_homepage.png` (nmap.org homepage), `screenshots/task03_scanme_notice.png` (scanme.nmap.org authorization notice).

Task 4 — Perform a Whois Query

Purpose: Find out who registered the domain and which infrastructure it is publicly associated with.

Required scope: query the **parent registered domain** (`nmap.org`), not the subdomain. Whois records exist at the registered-domain level.

Output file: `target_whois.txt` (in `~/recon_lab/`).

Verification step: the file contains the strings `Registrar:`, `Name Server:`, and at least one date field (`Creation Date:`, `Created:`, or equivalent).

Information to record:

- Registrar.
- Name servers.
- Creation date.
- Updated date.

- Expiration date.
- Any visible registrant fields (organization, country, abuse contact).

Evidence to submit: `target_whois.txt` , `screenshots/task04_whois.png` .

Task 5 — Conduct DNS Reconnaissance with Nslookup

Purpose: Practice DNS lookup against the system's default resolver.

Required mode: use `nslookup` in **interactive mode** (enter the tool, then issue the query at its prompt).

Output file: none.

Verification step: the screenshot shows a `Server:` line (the default resolver) and an A-record `Address:` line for the target.

Information to record:

- Default DNS server.
- Target A record (IPv4 address).
- Whether the answer was labeled "Non-authoritative answer."

Evidence to submit: `screenshots/task05_nslookup.png` .

Task 6 — Query Authoritative DNS Records

Purpose: Switch your DNS lookup to the domain's authoritative name server and query multiple record types directly.

Required coverage: within an `nslookup` session, switch the server to one of the authoritative name servers and query all of:

- **A** record for `nmap.org` .
- **MX** records for `nmap.org` .
- **NS** records for `nmap.org` .
- **CNAME** record for `www.nmap.org` .
- **SOA** record for `nmap.org` .

Output file: none.

Verification step: the screenshot shows the default-server switch to an authoritative name server and answers for each of the five record types. CNAME may legitimately return no answer — that is a valid finding.

Information to record:

- A record.
- MX records (hostname and priority of each).
- NS records.
- CNAME record (or "none returned").
- SOA record (primary NS, responsible mailbox, serial number).

Evidence to submit: `screenshots/task06_nslookup_auth.png` .

Task 7 — Capture DNS Information Using Dig

Purpose: Repeat the DNS queries non-interactively and capture all results into a single file using shell redirection.

Required coverage: capture five separate queries into one file:

1. Default query for `nmap.org` .
2. Default query for `scanme.nmap.org` .
3. MX query for `nmap.org` .
4. NS query for `nmap.org` .
5. SOA query for `nmap.org` .

The first query creates the file (overwrite); each subsequent query appends to it.

Output file: `target_dns.txt` (in `~/recon_lab/`).

Verification step: running `grep -c "ANSWER SECTION" target_dns.txt` returns at least 4. The file contains the keywords `MX` , `NS` , and `SOA` .

Information to record:

- That all five queries were captured.
- Any differences between the dig results and the nslookup results from Task 6.

Evidence to submit: `target_dns.txt` , `screenshots/task07_dig.png` .

Task 8 — Practice Safe Search Engine Reconnaissance

Purpose: Learn how search-engine operators reveal public information about an organization.

Required coverage: run at least four searches, each using a different operator, against approved demo targets (`example.com` , `example.org` , or others listed on your lab portal). Operators to demonstrate at least once across your four searches:

- `site:`
- `filetype:`
- `intitle:`
- A quoted exact phrase
- The minus (`-`) exclusion operator

Scope reminders: - Do **not** search for passwords, credentials, leaked databases, or any real person. - Do **not** run search operators against `nmap.org` or `scanme.nmap.org` — use only the approved demo targets.

Output file: none.

Verification step: screenshots clearly show the search query (in the URL or search box) and the result list.

Information to record (for each of the four searches):

- The exact query used.
- The operator(s) demonstrated.
- The type of result returned (web page, PDF, image, etc.).
- The relevant public information visible.
- The security value — what could a defender or attacker learn from this technique applied to a real organization?

Evidence to submit: `screenshots/task08_search1.png` , `screenshots/task08_search2.png` .

Task 9 — Use an Approved Website OSINT Lookup Tool

Purpose: Practice passive fingerprinting using an approved OSINT lookup site.

Required scope: use only the lab-approved website technology / hosting lookup tool (URL on your lab portal). Query `nmap.org` .

Output file: none.

Verification step: the screenshot shows the OSINT tool's results page for `nmap.org` .

Information to record (whichever fields the tool displays):

- Hosting provider.
- Server technology.
- First-seen date.
- Related hostnames.
- Public-facing technologies (JS libraries, analytics, frameworks).

Evidence to submit: `screenshots/task09_osint.png` .

Task 10 — Perform Public Data Awareness Review

Purpose: Build awareness of OSINT exposure patterns using a safe fictional dataset.

Source: the pre-staged fictional dataset in `~/recon_lab/demo_people_data.txt` .

Required analysis: identify at least **four categories** of public information present in the dataset. For each category:

- Explain why it matters (the misuse risk — phishing, pretexting, password guessing, etc.).
- Suggest at least one practical control that reduces the exposure.

Scope reminder: all data in `demo_people_data.txt` is fictional. Do not attempt to look up real people who share names with the dataset.

Output file: none (your analysis goes in the report).

Verification step: your report's *Public data awareness* section contains at least four data categories, each paired with a misuse risk and a reduction strategy.

Information to record:

- Categories of public information identified.
- Why each category matters.
- How an organization can reduce exposure.

Evidence to submit: content in the report; no separate file.

Task 11 — Organize Reconnaissance Evidence

Purpose: Hand off evidence the way a real analyst would.

Expected structure (every required file in the right location):

```
~/recon_lab/
├─ target_info.txt
├─ target_whois.txt
├─ target_dns.txt
├─ recon_report.docx      (or .pdf / .md)
├─ screenshots/
│   ├── task01_ip.png
│   ├── task02_ping.png
│   ├── task03_homepage.png
│   ├── task03_scanme_notice.png
│   ├── task04_whois.png
│   ├── task05_nslookup.png
│   ├── task06_nslookup_auth.png
│   ├── task07_dig.png
│   ├── task08_search1.png
│   ├── task08_search2.png
│   ├── task09_osint.png
│   └─ task11_evidence.png
```

Output file: organized `~/recon_lab/` directory.

Verification step: a recursive directory listing of `~/recon_lab/` shows every required file in its expected location.

Information to record: none (structural verification only).

Evidence to submit: `screenshots/task11_evidence.png` — a screenshot of the recursive directory listing.

Task 12 — Document and Correlate Reconnaissance Findings

Purpose: Convert raw output into a written analyst report that a senior analyst could act on.

Required content (per the report template in Section 10):

- Findings from all tasks recorded in the appropriate sections of the template.
- A *Key risks* section that ties each significant finding to a "so what" — why it matters.
- A *Recommended next steps* section that gives concrete suggestions for:
- **Vulnerability management** (e.g., now that we know the mail provider is Google Workspace, DMARC/SPF/DKIM posture becomes a relevant control to review).
- **Third-party risk review** (e.g., Linode hosts `scanme.nmap.org`, Linode DNS hosts `nmap.org`, Google handles mail — all third parties whose security postures matter to anyone consuming the Nmap Project's outputs).

Output file: `recon_report.docx` or `recon_report.pdf` in `~/recon_lab/`.

Verification step: every section of the report template is filled in; no placeholder underscores remain.

Information to record: all findings from Tasks 1–10, structured per the report template.

Evidence to submit: `recon_report.docx` (or `.pdf`).

8. Deliverables

Submit the following from your `~/recon_lab/` folder:

1. `target_info.txt`
2. `target_whois.txt`
3. `target_dns.txt`
4. `recon_report.docx` (or `recon_report.pdf`).
5. The `screenshots/` folder containing all 12 named screenshots: - `task01_ip.png` - `task02_ping.png` - `task03_homepage.png` - `task03_scanme_notice.png` - `task04_whois.png` - `task05_nslookup.png` - `task06_nslookup_auth.png` - `task07_dig.png` - `task08_search1.png` - `task08_search2.png` - `task09_osint.png` - `task11_evidence.png`

9. Submission Instructions

1. Package your evidence into a single archive:

```
bash cd ~ zip -r recon_lab_submission.zip recon_lab/
```

2. Log in to the bootcamp learning portal.
3. Open the **Performing Reconnaissance** lab page.
4. Click **Submit Lab** and upload `recon_lab_submission.zip`.
5. Confirm the submission status changes to "Submitted — awaiting review."

If the portal is unavailable, upload the zip to the assigned shared folder and notify your instructor by message.

10. Learner Report Template

Copy the structure below into `recon_report.docx` (or `.md`) and complete every section.

Reconnaissance Report

Analyst: *Your name*

Date: *Today's date*

Assigned target: `scanme.nmap.org` (parent domain `nmap.org`)

1. Target information

- Target hostname:
- Parent domain:
- Target IP address (from ping/dig):
- Authoritative DNS servers:
- Operator authorization notice (summarized):

2. Kali network identity

- Kali IP address:
- Interface:
- Subnet:

3. Connectivity results

- Target reached (Y/N):
- Packets transmitted:
- Packets received:
- Packet loss (%):
- Average round-trip time:

4. Website findings

- Project / site display name:
- Page types observed on `nmap.org`:
- Visible technologies:
- Authorization notice summary (from `scanme.nmap.org`):
- Other public-facing information:

5. Whois findings (for `nmap.org`)

- Registrar:
- Name servers:
- Creation date:
- Updated date:
- Expiration date:
- Other relevant fields:

6. DNS findings

- Default resolver (from nslookup):
- A record:
- MX record(s):
- NS record(s):
- CNAME record (or "none returned"):
- SOA record:
- Notes on differences between nslookup and dig results:

7. Search reconnaissance findings

- Query 1 — operator / result type / what it revealed:
- Query 2 — operator / result type / what it revealed:
- Query 3 — operator / result type / what it revealed:
- Query 4 — operator / result type / what it revealed:

8. OSINT lookup findings (for nmap.org)

- Hosting provider:
- Server technology:
- First-seen date:
- Related hostnames:
- Other technologies:

9. Public data awareness observations

- Categories of public information identified:
- Why each category matters:
- Suggested ways to reduce exposure:

10. Key risks

- Risk 1 — finding and why it matters:
- Risk 2 — finding and why it matters:
- Risk 3 — finding and why it matters:

11. Recommended next steps

- Next step 1 (vulnerability management):
- Next step 2 (third-party risk review):
- Next step 3 (additional escalation, if any):

12. Evidence list

- [target_info.txt](#)
- [target_whois.txt](#)
- [target_dns.txt](#)
- [screenshots/](#) (list each file)
- [recon_report.docx](#)